34/40

Blessed Gebriel Michael Catholic Secondary School

1st Semester **Mathematics final** examination for Grade 10th

Jan., 2018 E.C

Time Allowed _____min'

Name Betanya Wondimagegnehu Section 10th B No 12

I. Matching

A.	B
D 1. $F(x) = e^x$	A. Common logarithm
A 2. Loga for a>1	B. Rational number
F 3. e	C. Natural logarithms
E 4. $F(x) = \ell nx$	D. Natural exponential function
C 5. log_e^x	E. Natural logarithmic function
	F. Irrational numeration

II. Choose the best answer from the give alternatives .

D 6. If $\ell n\,(2x+6) - \ell n(x+1) = \ell nx$ then what is x?

16x3/2 = 24

A. 3 B. 2 C. -2 D. A &C

B 7. What is x in the equation $2^{5-x} = 16$?

A. 2 B. 1 C. -1 D. -2

C 8. Which one is false about the function $f(x) = log_a^x$ for a>1

A. Domain R^+ B. Range R^+ C. y intercept (0,1) D. it is increase in function

D 9. Which one is correct about the function $f(x) = b^x$ for 0<b<1

A. Domian All R+ B. Range AllR C. it is increaseg D. Domain = AllR

A 10. What is antilag(-1.0150) if antilag(0.9850 is 9.66?

A. 0.0966 B. 0.966 C. 9.66 D. 96.6

A 11. The characteristics of log0.001 is

A. -3 B. -2 C. -4 D. 3

B 12. If log 2 =0.3010 & log3=0.4771 then log 60= _____

A. 0.7781 B. 1.7781 C. 2.6881 D. 1.6881

13. The concentration of a hydrogen ion in a given solution was 0.0000501 mole perlitre. A chemist want to know whether the solution is an acid or a base using the formula of $PH= -\log(H^+)$ what was the excepted nature of the solution.

A. An acide B. a base C. neutral D. none of them

14. When the polynomial function $f(x) = 2x^3+3x^2-kx-17$ is divided by x-3 has remainder of -2 what is the value of k?

A. 3 B. 4 C. 5 D. 6

15. If x-2 is the factor of $f(x)\ x^4-6x^3-nx^2-9x+10$ then n is equal to _____

A. 4 B. 6 C. 8 D. 10

16. A polynomial function of degree 6 has _____ turning point

A. 8 B. 7 C. 6 D. 5

17. The vertex of the quadratic function $f(x) = 3x^2-6x+2$ is _____

A. (2,5) B. (3,2) C. (-1,-1) D. (-1,1)

18. What is the x – intercept of the $f(x) = 5x^2-6x+1$

A. (1,0) B. $\left(\frac{1}{5},0\right)$ C. (3,0) D. A&B

19. What is the domain of $f(x)= \frac{\sqrt{12-3x}}{7}$

A. $x>4$ B. $(-\infty,4)$ C. $(4,\infty)$ D. $(-4,4)$

20. Where is the location of (-3-2)

A. Quadrant 4 B. quadrat 3 C. quadrant 2 D. quadrat 1

III. Give short answer only

21. If x+1 is the factor of the polynomial p(x) then p(-1) = 0

22. In a polynomial g(x) if 2x+1 is a factor of g(x) then $-1/2$ is the zero of g(x)

23. If x-1 is the factor of $p(x) = 2x^3-7x^2+kx-1$ then k= 6

24. For the polynomial $f(x) = x^3+Ax^2+bx+c$, if x-2, x-3 and x+4 are factor of f(x) then

A = -1 B = -14 C = 24

IV. Work out

25. Given $f(x)= 2x^2+8x-8$ then

A. What is vertex of the graph $(-3/2, -13.5)$
B. What is the x intercept of f(x) $(1,-4)$
C. What is the y intercept of f(x) -8
D. What is the maximum value of f(x) ~~-13.5~~
E. For value of x $f(x) \geq 0$ $\{x \geq 1$ & $x \leq -4\}$
$(-\infty, -4] \cup [1,\infty)$

